University of Basel

Compositional Processes as Graph
Semantic Modelling and Terminology in the Anton Webern Gesamtausgabe (AWG)

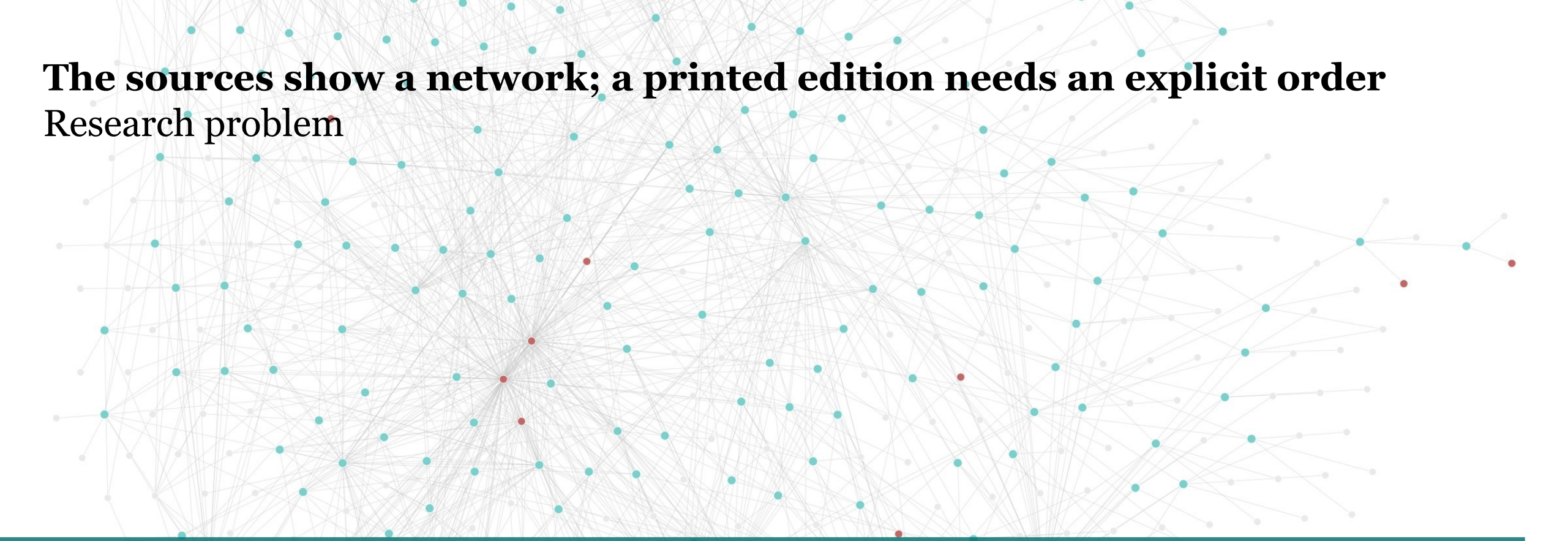


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Comparative Perspectives on Genetic Criticism in Music
International Conference of the IMS Study Group
Rome, German Historical Institute, 16–18 November 2026

The sources show a network; a printed edition needs an explicit order

Research problem



How can a digital edition represent non-linear compositional processes – and keep philological claims as open as the evidence demands?

Source: AWG online edition, op. 25, graph view of all triples (development version, screenshot 11 Nov 2026)

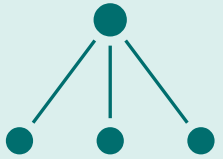
The AWG graph separates what the data imply from what editors assert

The AWG and its graph

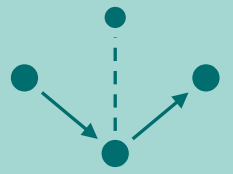
draft ontology v0.1

edition.anton-webern.ch

generated layer
derived from the edition's structured data



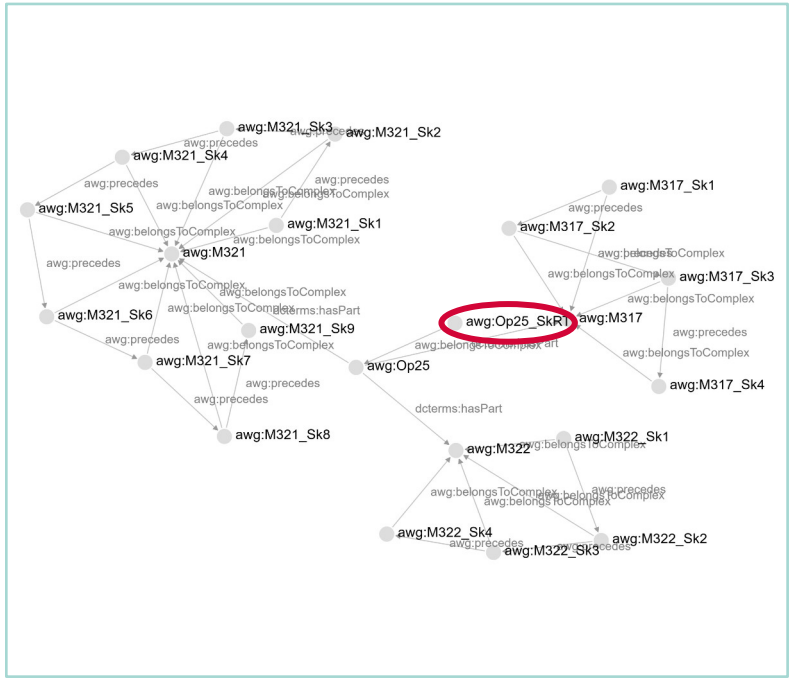
curated layer
hand-modelled editorial assertions



Queried with SPARQL in the app, exported as Turtle. No reasoning.

In op. 25, the layers show where editorial judgment begins

Live demo: op. 25



1 generated
three islands, one per song



2 curated
one chain across the songs



3 both
the chain crosses the opus order

Source: AWG online edition, op. 25, graph view filtered by SPARQL (development version, screenshots 11 Nov 2026)

The categories come from Webern's documents, not from theory

Terms from the material

sketch
`Skizze`

Sk1, Sk2, Sk2.1: the editors' numbering in the source descriptions

version
`Textfassung`

TF1, TF2: recognisable states of a song in the sources

paratext
`Paratext`

title, dating, instruments: the categories of the source descriptions

row table
`Reihentabelle`

Webern's sheet of row forms, a source of its own in op. 25

edition complex
`Editionskomplex`

what is edited together, named by opus or M number

op. 25 =

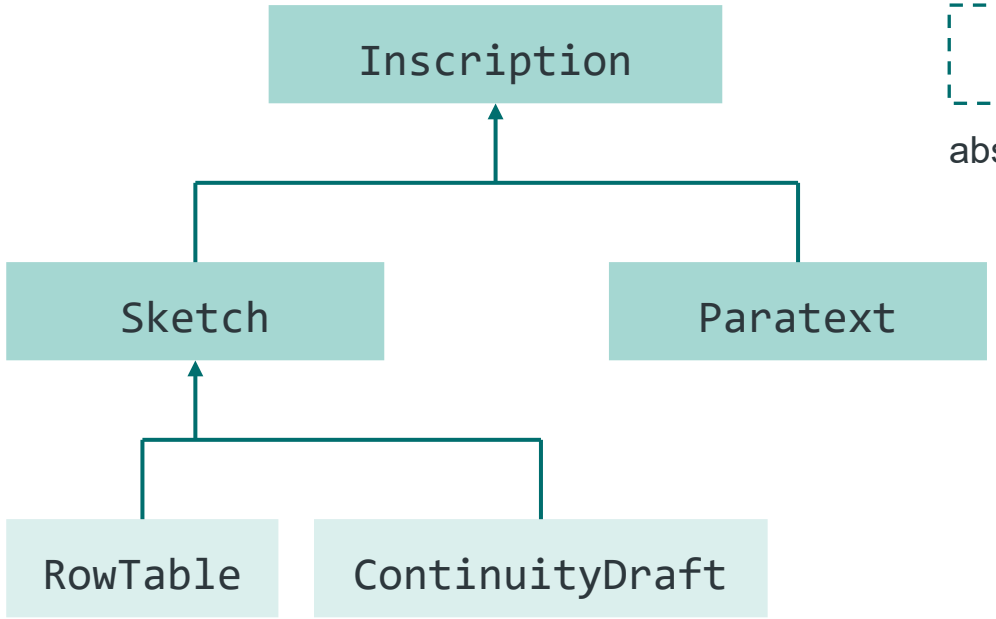
M 317	M 322	M 321
song I	song II	song III

a composite edition complex: parts in the order of the opus

Editorial terms become classes; uncertainty becomes a tag

From term to class

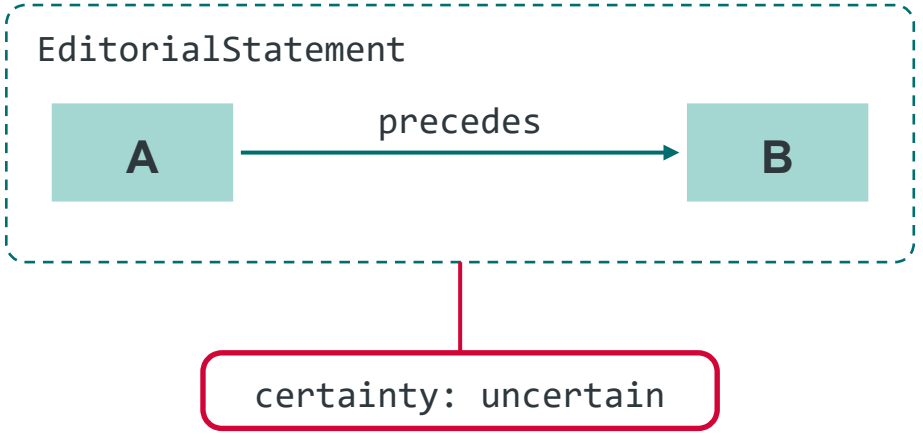
classes



`TextVersion`

abstract, not the edited sheet

uncertainty



a tag on the statement, not a relation of its own;
without a tag, the statement counts as certain

Classes of the draft ontology v0.1, prefix `awg:` omitted. **Not modelled yet:** “study composition”, “ink draft”.

Two relations carry the genetic order: a sequence, or mutual influence

Genetic relations

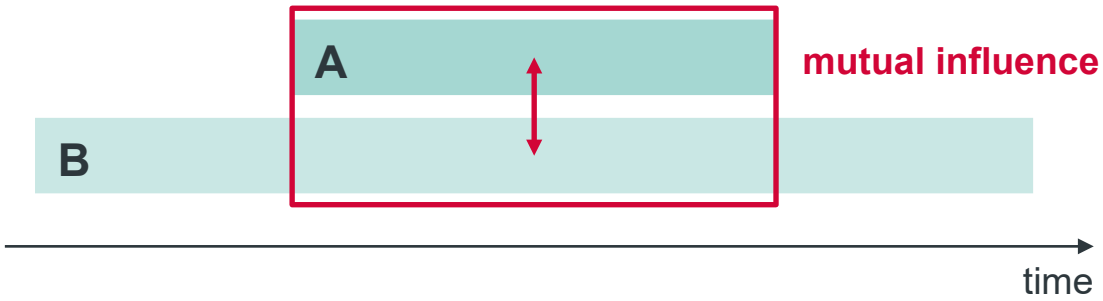
`precedes`



an order can be shown: A begins before B starts (overlap allowed)

cf. CIDOC CRM P176 “starts before the start of”

`concomitates`



A and B influence each other; which came first cannot be decided

cf. CIDOC CRM P174 “starts before the end of” (the overlap, not the influence)

evidence: subproperties of `precedes`

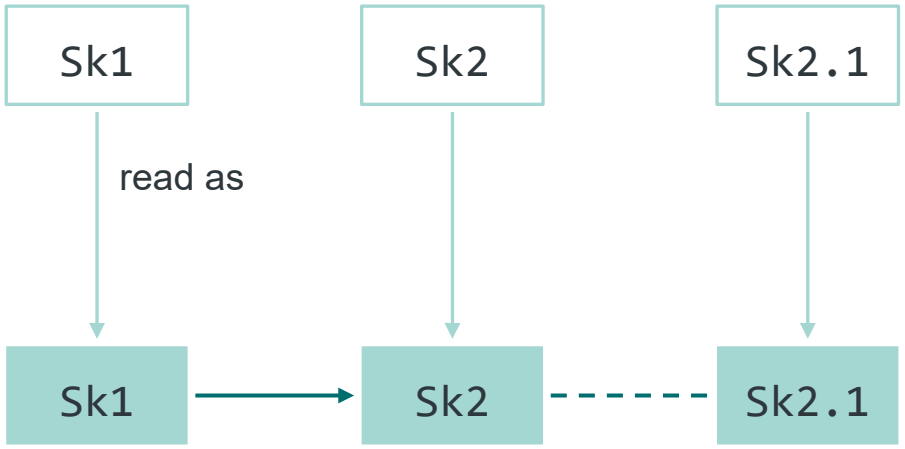
`precedesScripture`	`precedesContent`
page, overlay, writing instruments	musical content

The model makes explicit what editorial practice leaves implicit

Semantic shifts

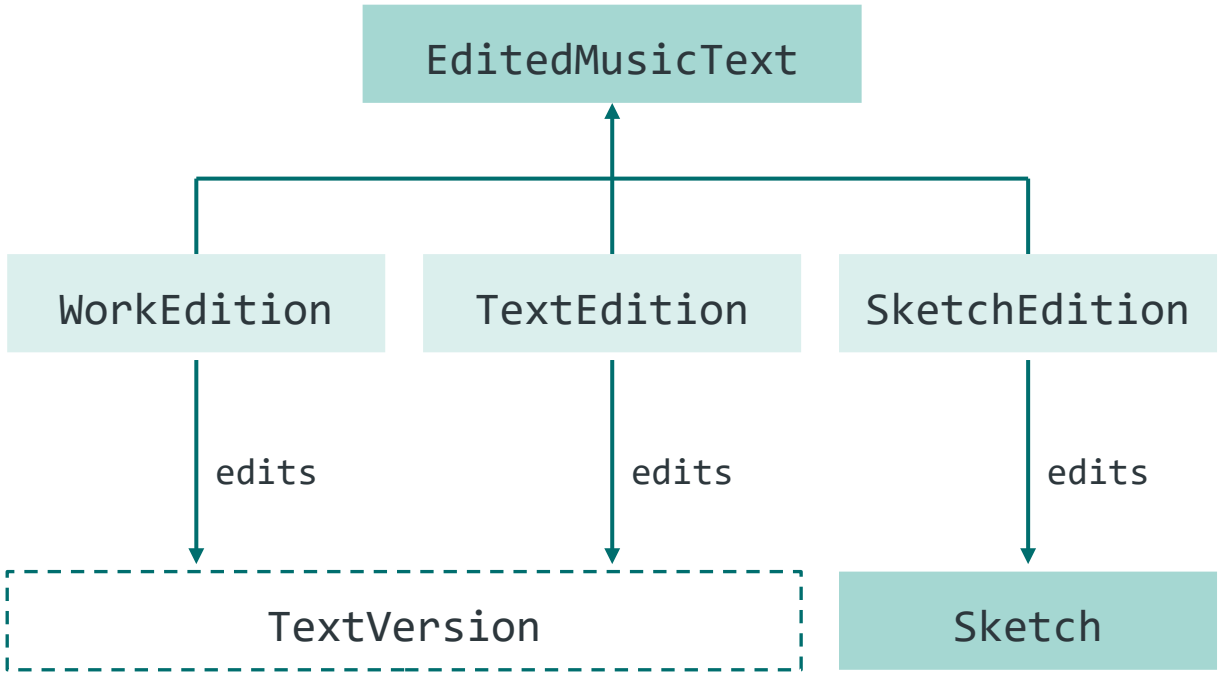
numbering becomes process

source description: labels



graph: genetic relations

three editions, one superclass



what the edition makes, linked to what it edits

The model transfers, the pipeline does not

What transfers

portable	
`Inscription`	what is written and ordered
`Document`	the physical object
`Folio`	one page side
`Placement`	where on the page: staves, measures
`precedes`	genetic order, overlap allowed
`concomitates`	mutual influence, no order

tied to the AWG

the pipeline
the generator reads the AWG's own data

the Moldenhauer catalogue
M numbers name the complexes

Transfer to other repertoires is plausible but untested.

Where claims stay open, the graph pays in queryability

Frictions: model vs. data

model lags data		data lags model
staves free text: “1c–3c”		**sketchbooks** per complex: SB3 as two excerpts
measures free text: “{1A}, {1B}, 1–2, 3A”	⟷	Sk## **placeholders** `warning: placeholder` 29 sketches without stable identifier
writing instruments missing: pencil, ink, crayon		**incomplete dates** `warning: incompleteDate` “M. am 30. IX. …”, no year

Each gap is information.

Standards serve as alignments, not as the model

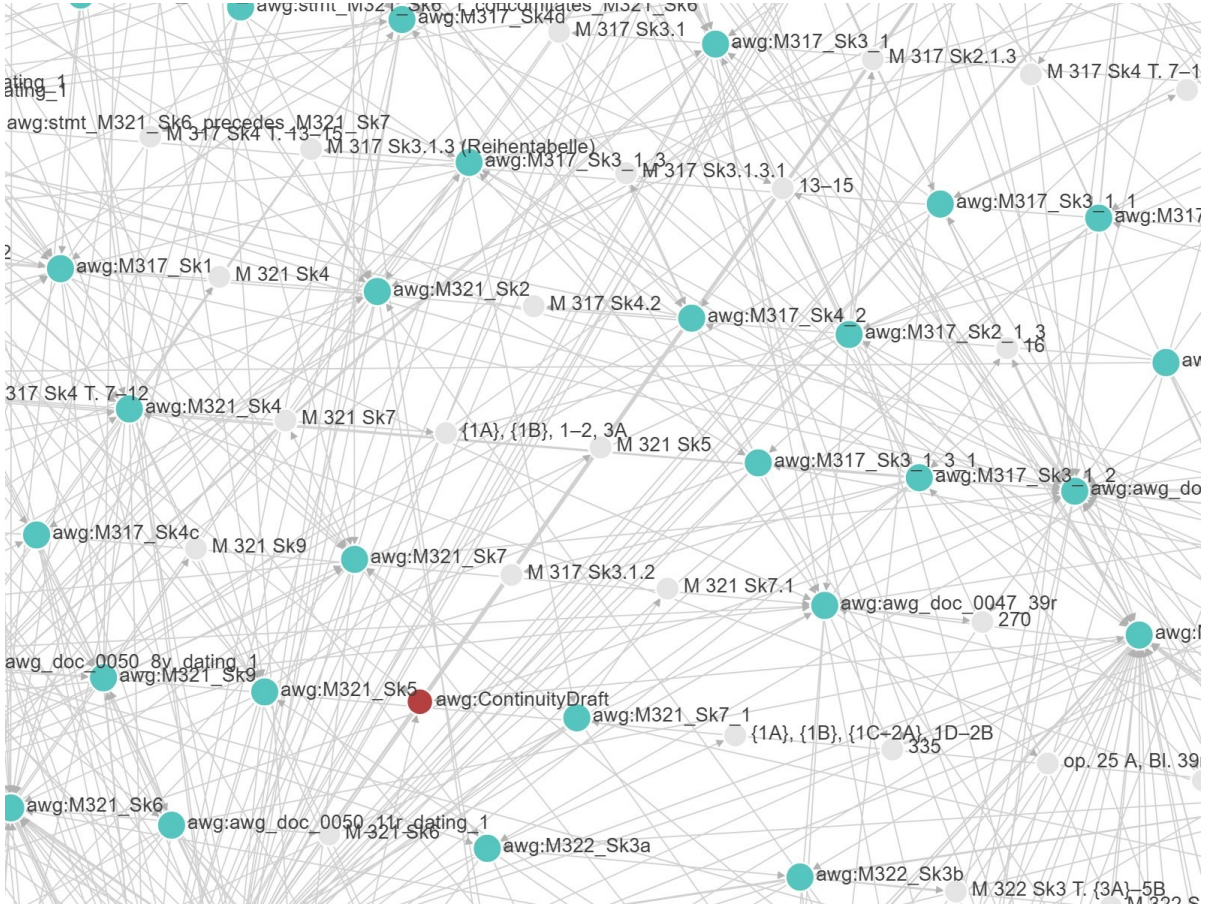
Standards and alignments

The sketch is content and object at once – which LRMoo separates (F2 Expression vs. F5 Item).

standard	in the AWG model	link
LRMoo	edition complex ≈ F1 Work · version, sketch ≈ F2 Expression · document ≈ F5 Item	`rdfs:seeAlso`
CIDOC CRM	temporal reading: precedes ≈ P176, concomitates ≈ P174	`rdfs:seeAlso`
Dublin Core terms, SKOS, PROV, `rdf:Statement`	reused outright: parts, concept schemes, attribution, statements	`used directly`
MEI	not integrated yet	–

The mismatch between model and data is the analytical gain

Conclusion and outlook



Source: AWG online edition, op. 25, graph view of all triples (development version, screenshot 11 Nov 2026)

> Layers show where editorial judgment begins.

> Warnings show where the data disagree.

> Model and data are written side by side: concomitant.

Next steps: named graphs for ink and pencil layers, persons, archives

If model and data shape each other, the graph is also our construction

Closing question

Whose genetic order does the graph show – Webern's or ours?

How much of Webern's compositional process survives translation into shared standards?

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Thank you

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